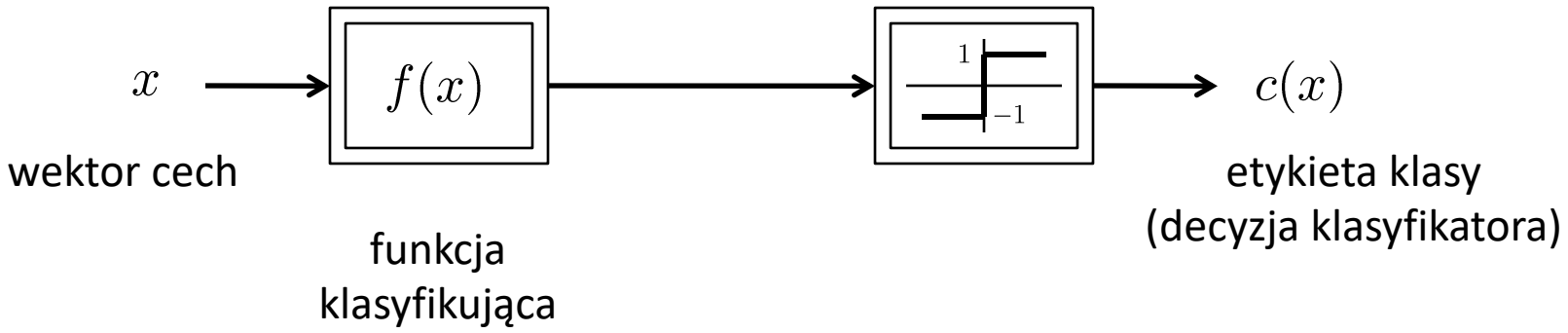
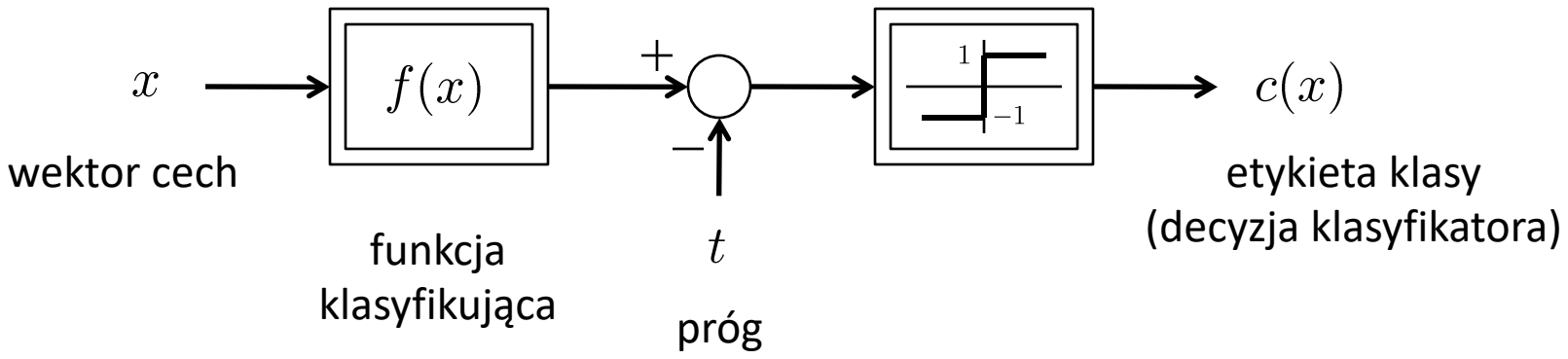
Metody oceny skuteczności systemów rozpoznających

Krzysztof Fujarewicz

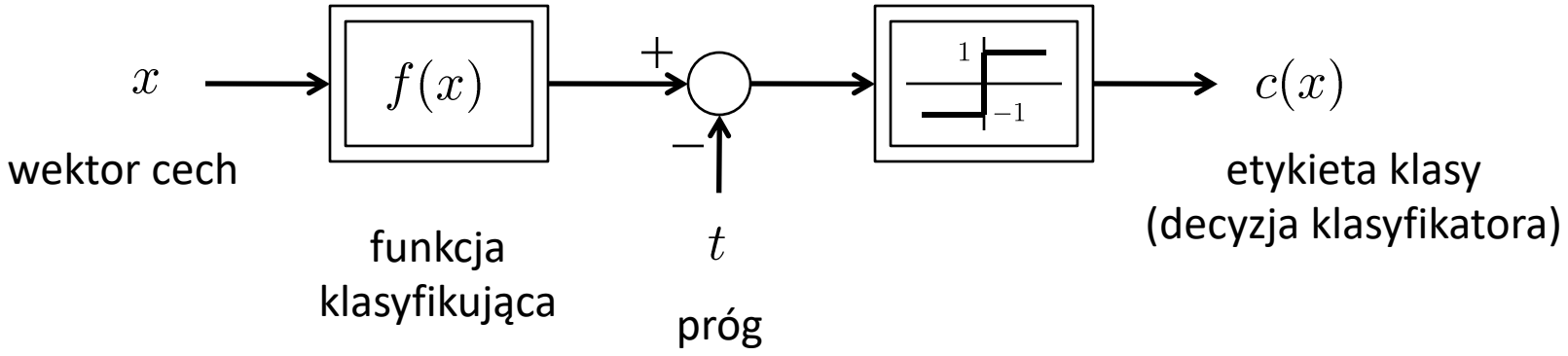
Ogólna struktura klasyfikatora

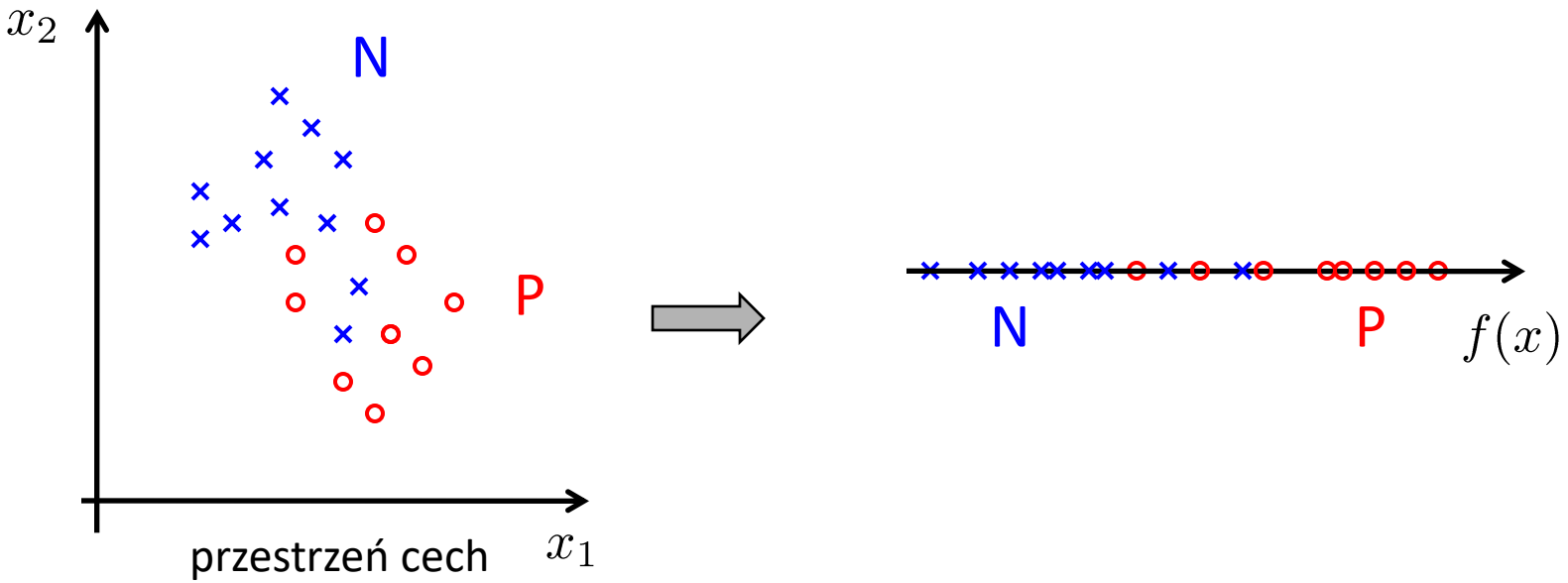


Ogólna struktura klasyfikatora

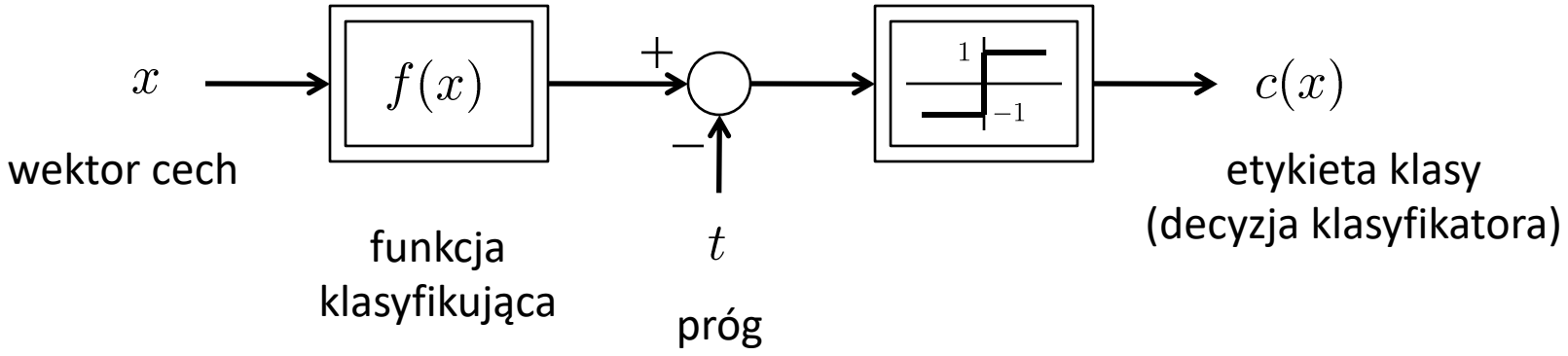


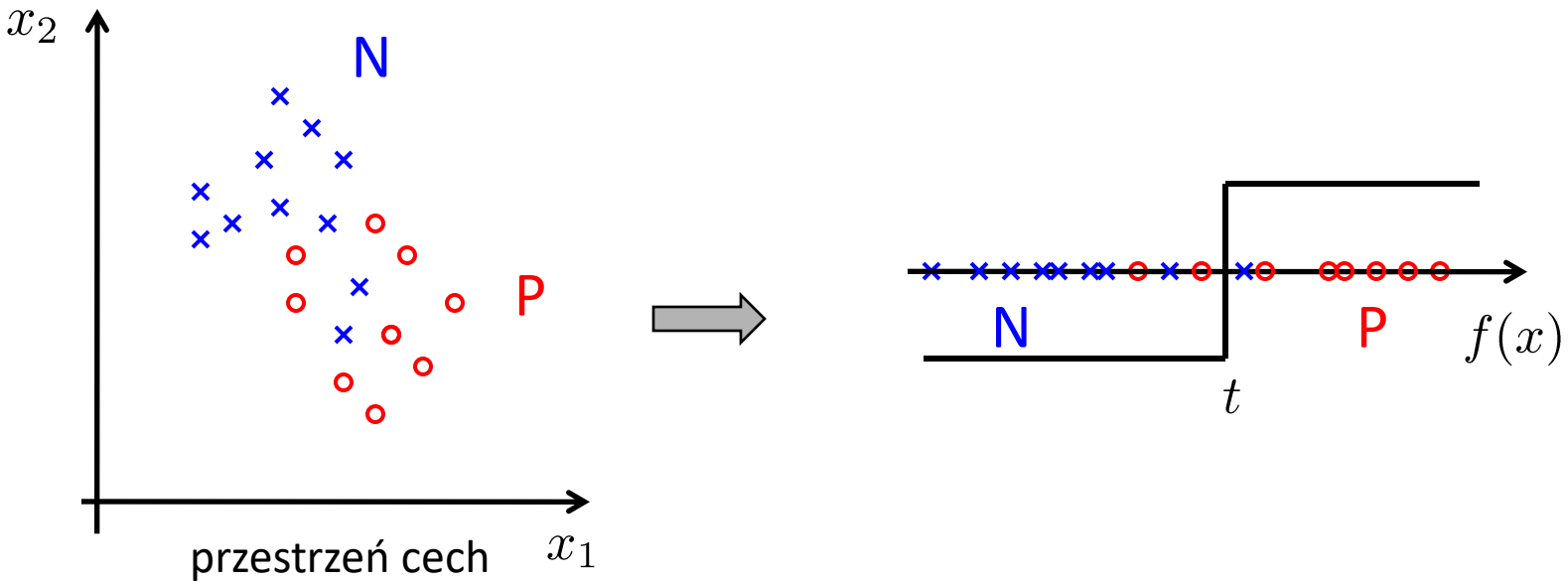
Ogólna struktura klasyfikatora



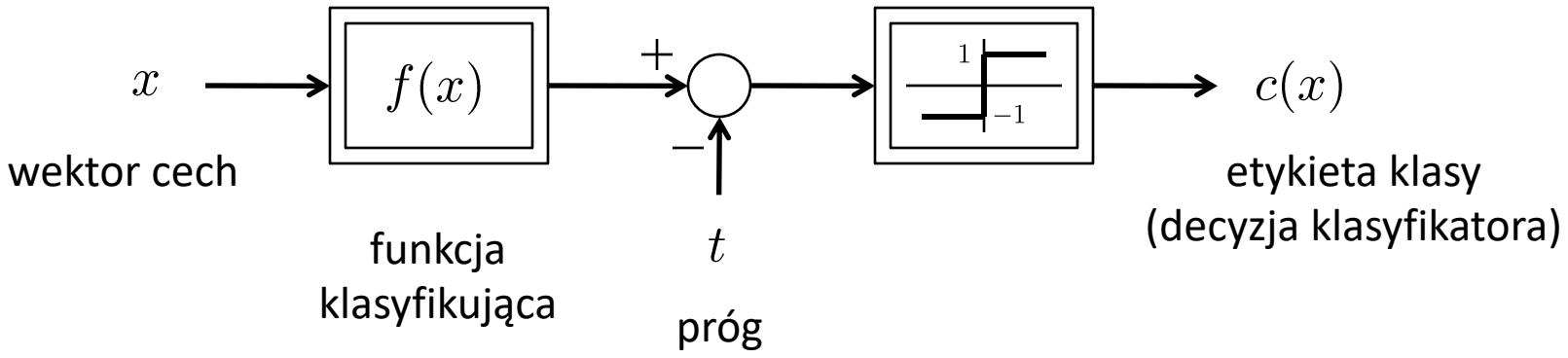


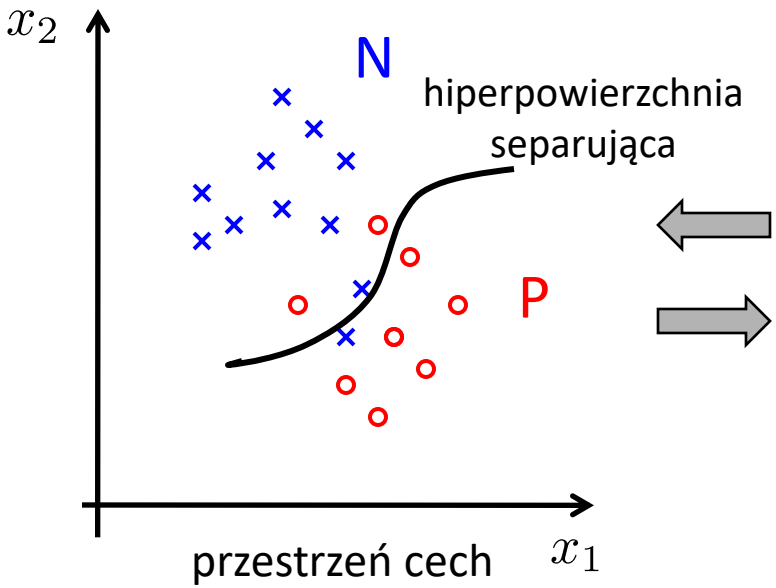
Ogólna struktura klasyfikatora

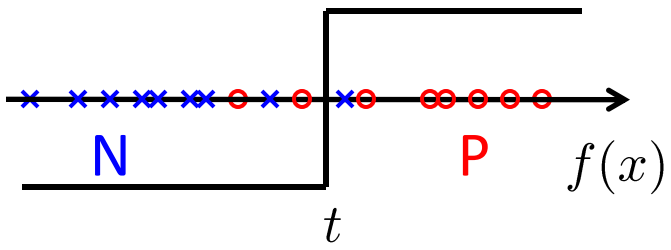




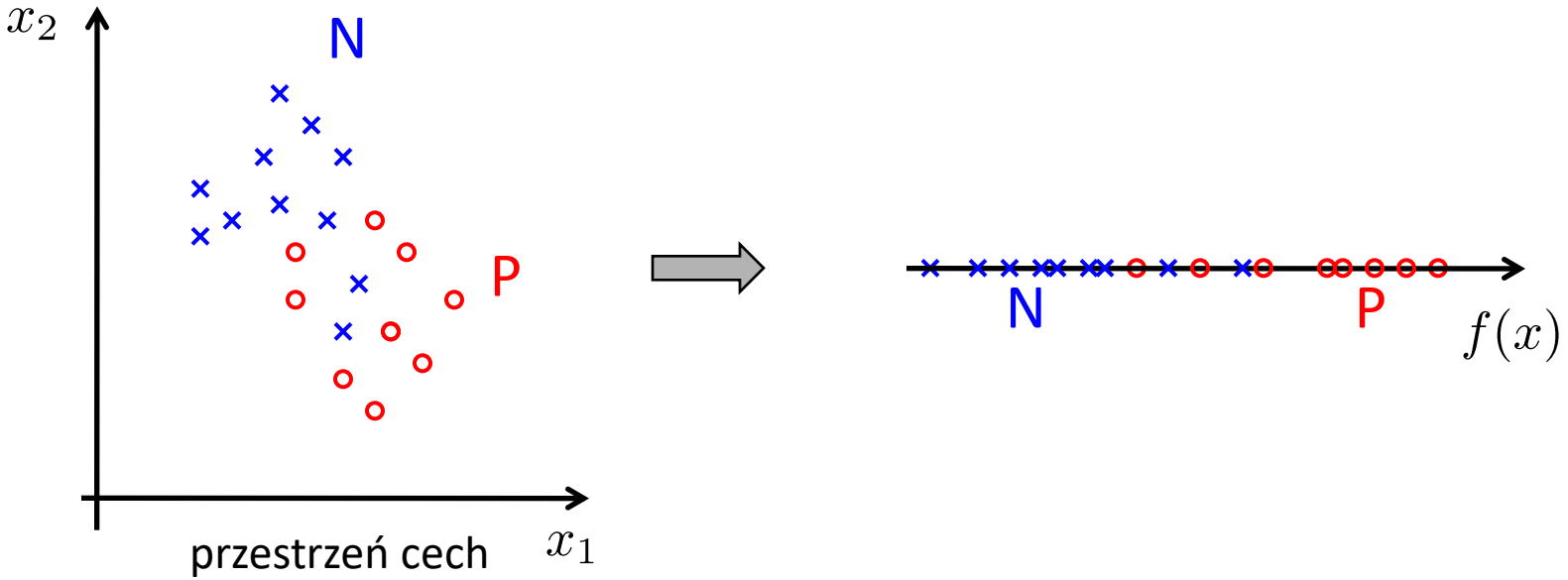
Ogólna struktura klasyfikatora





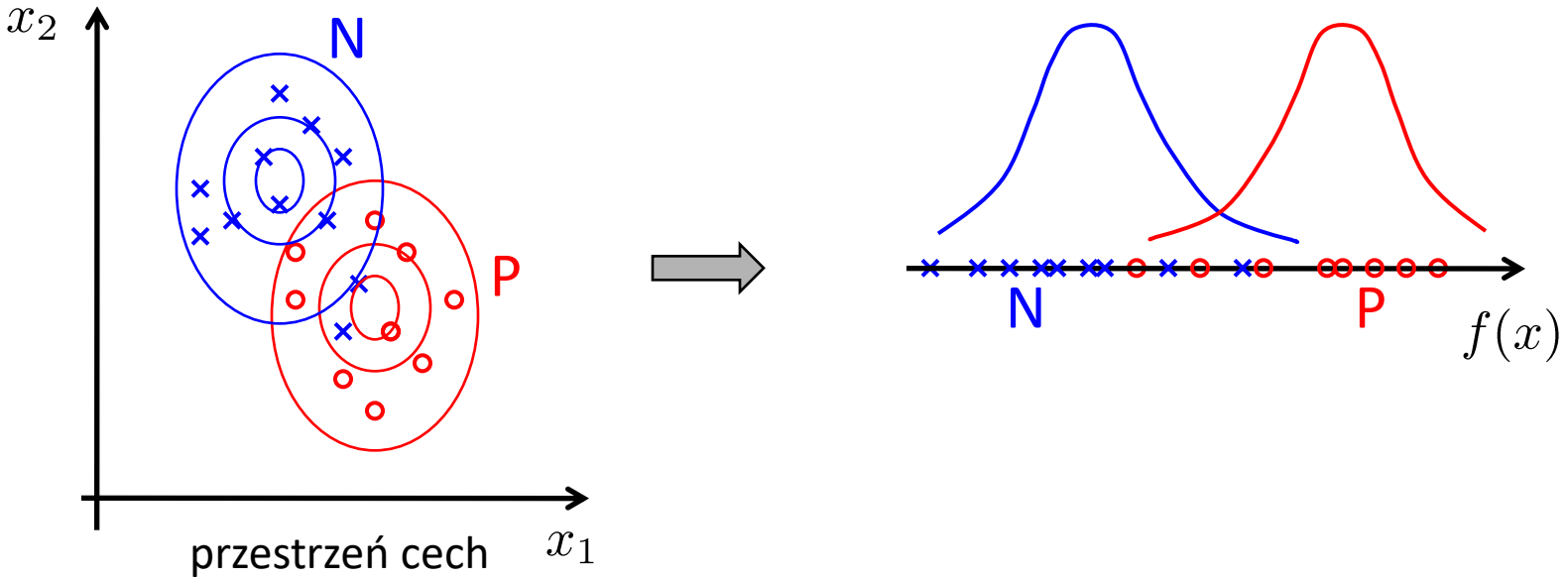


Ogólna struktura klasyfikatora



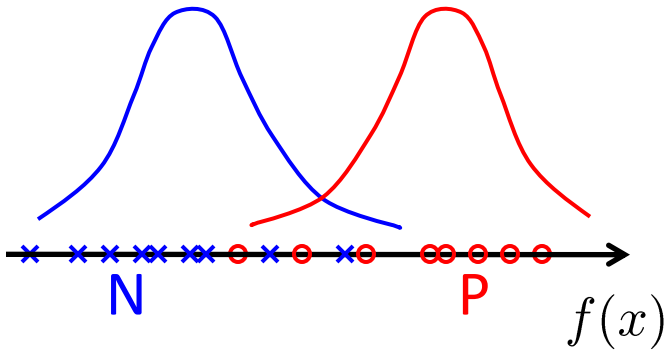
Zbiory danych P i N są próbami z populacji o nieznanych rozkładach

Ogólna struktura klasyfikatora

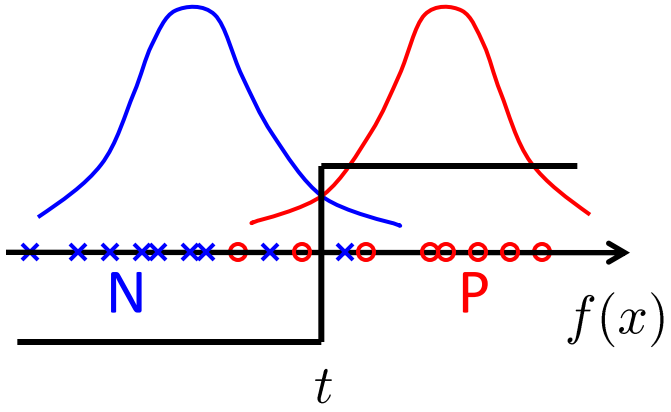


Zbiory danych P i N są próbami z populacji o nieznanych rozkładach

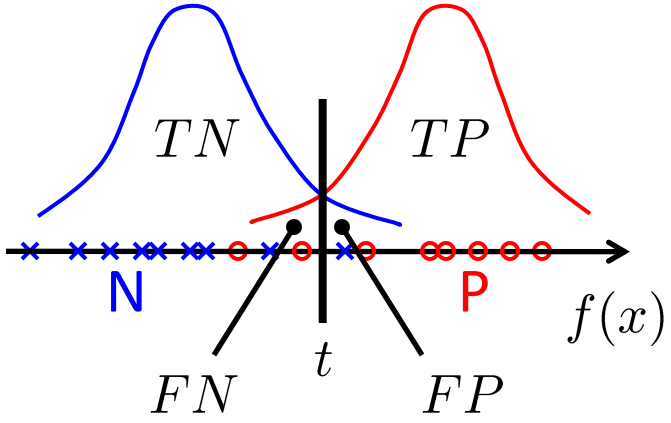
Tablica kontyngencji



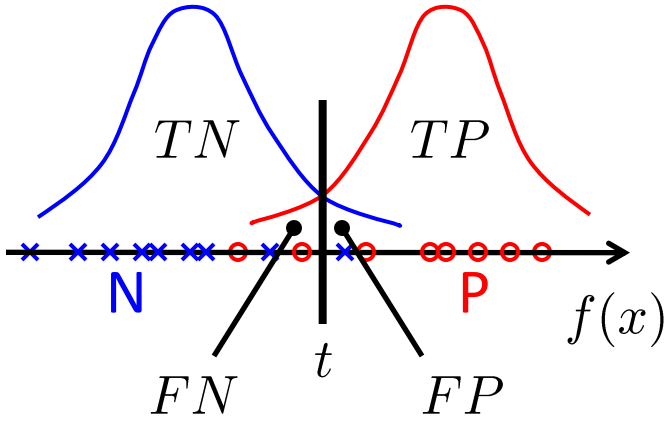
Tablica kontyngencji



Tablica kontyngencji

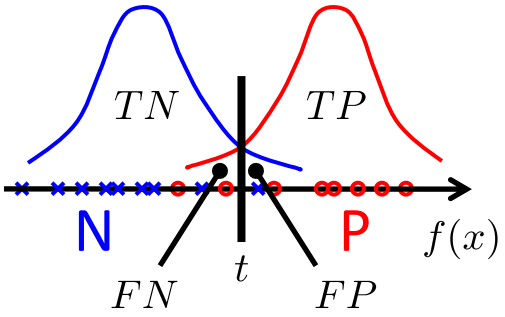


Tablica kontyngencji



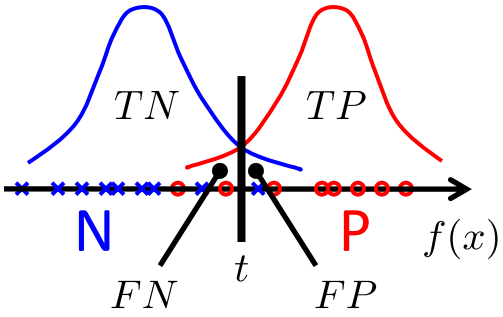
	Decyzja klasyfikatora	
Wiedza eksperta	$\mathrm{P}_{\mathrm{klas}}$	$\mathrm{N}_{\mathrm{klas}}$
$\mathrm{P}_{\mathrm{exp}}$	TP	FN
$\mathrm{N}_{\mathrm{exp}}$	FP	TN

Wskaźniki skuteczności klasyfikatora



	Decyzja klasyfikatora	
Wiedza eksperta	$\mathrm{P}_{\mathrm{klas}}$	$\mathrm{N}_{\mathrm{klas}}$
$\mathrm{P}_{\mathrm{exp}}$	TP	FN
$\mathrm{N}_{\mathrm{exp}}$	FP	TN

Wskaźniki skuteczności klasyfikatora

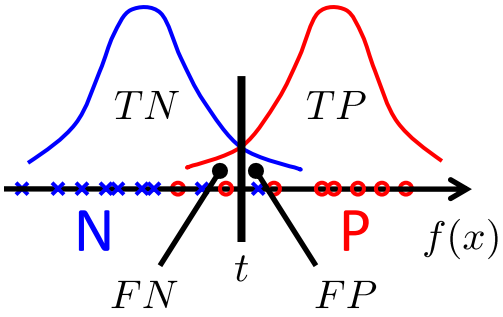


	Decyzja klasyfikatora	
Wiedza eksperta	$\mathrm{P}_{\mathrm{klas}}$	$\mathrm{N}_{\mathrm{klas}}$
$\mathrm{P}_{\mathrm{exp}}$	TP	FN
$\mathrm{N}_{\mathrm{exp}}$	FP	TN

Błąd klasyfikatora

$$Err = \frac{FP+FN}{TP+TN+FP+FN} \approx \Pr(F)$$

Wskaźniki skuteczności klasyfikatora

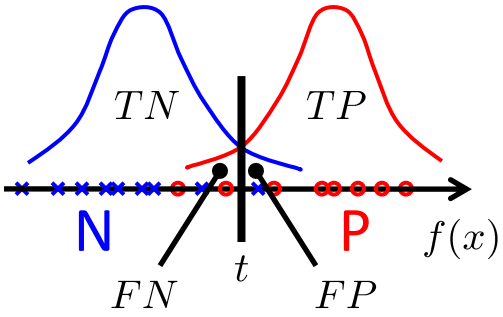


	Decyzja klasyfikatora	
Wiedza eksperta	$\mathrm{P}_{\mathrm{klas}}$	$\mathrm{N}_{\mathrm{klas}}$
$\mathrm{P}_{\mathrm{exp}}$	TP	FN
$\mathrm{N}_{\mathrm{exp}}$	FP	TN

Błąd klasyfikatora $$Err = \frac{FP+FN}{TP+TN+FP+FN} \approx \Pr(F)$$

Dokładność (precyzja) klasyfikatora $$Acc = \frac{TP+TN}{TP+TN+FP+FN} = 1 - Err \approx \Pr(T)$$

Wskaźniki skuteczności klasyfikatora



	Decyzja klasyfikatora	
Wiedza eksperta	$\mathrm{P}_{\mathrm{klas}}$	$\mathrm{N}_{\mathrm{klas}}$
$\mathrm{P}_{\mathrm{exp}}$	TP	FN
$\mathrm{N}_{\mathrm{exp}}$	FP	TN

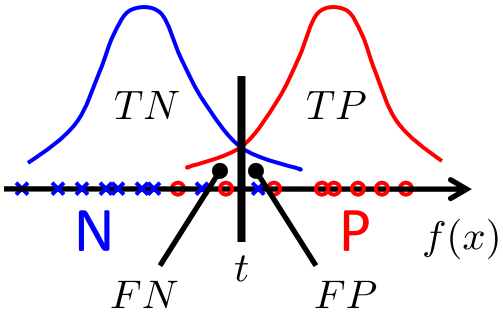
Błąd klasyfikatora $$Err = \frac{FP+FN}{TP+TN+FP+FN} \approx \Pr(F)$$

Dokładność (precyzja) klasyfikatora $$Acc = \frac{TP+TN}{TP+TN+FP+FN} = 1 - Err \approx \Pr(T)$$

Czułość klasyfikatora $$Sens = \frac{TP}{TP+FN} \approx \Pr(T|P_{exp})$$

Specyficzność klasyfikatora $$Spec = \frac{TN}{TN+FP} \approx \Pr(T|N_{exp})$$

Wskaźniki skuteczności klasyfikatora



	Decyzja klasyfikatora	
Wiedza eksperta	$\mathrm{P}_{\mathrm{klas}}$	$\mathrm{N}_{\mathrm{klas}}$
$\mathrm{P}_{\mathrm{exp}}$	TP	FN
$\mathrm{N}_{\mathrm{exp}}$	FP	TN

Błąd klasyfikatora $Err = \frac{FP+FN}{TP+TN+FP+FN} \approx \Pr(F)$

Dokładność (precyzja) klasyfikatora $Acc = \frac{TP+TN}{TP+TN+FP+FN} = 1 - Err \approx \Pr(T)$

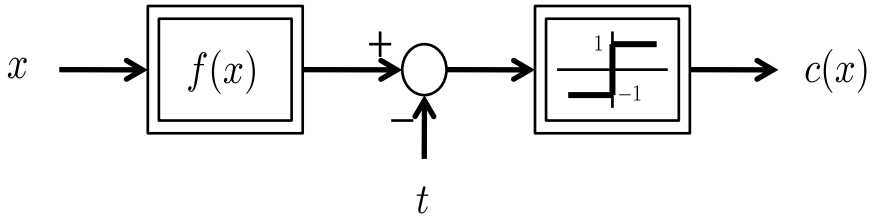
Czułość klasyfikatora $Sens = \frac{TP}{TP+FN} \approx \Pr(T|P_{exp})$

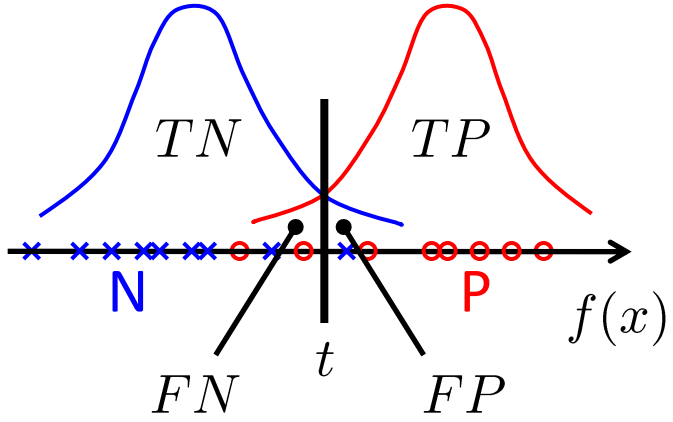
Specyficzność klasyfikatora $Spec = \frac{TN}{TN+FP} \approx \Pr(T|N_{exp})$

Dodatnia wartość predykcyjna $PPV = \frac{TP}{TP+FP} \approx \Pr(T|P_{klas})$

Ujemna wartość predykcyjna $NPV = \frac{TN}{TN+FN} \approx \Pr(T|N_{klas})$

Krzywa ROC
ang. Receiver Operating Characteristic



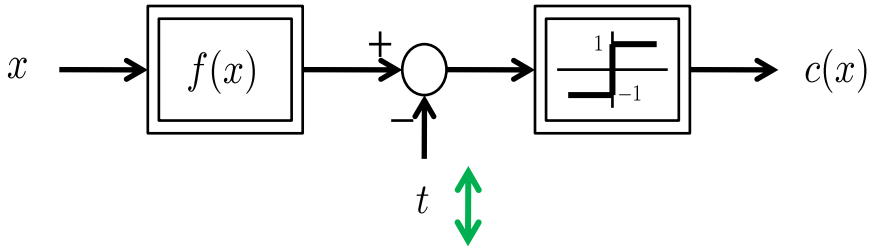


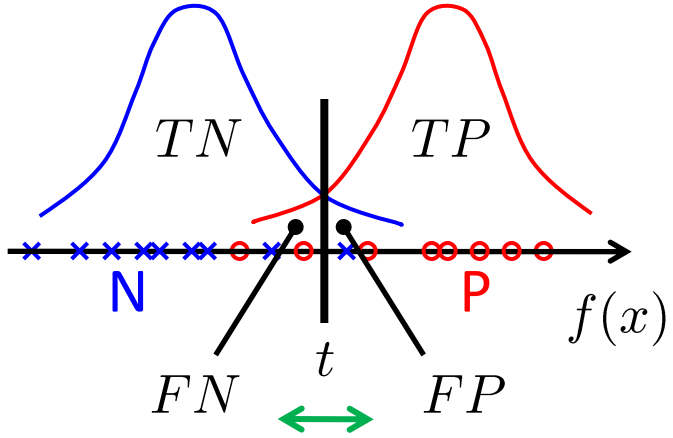
	Decyzja klasyfikatora	
Wiedza eksperta	$\mathrm{P}_{\mathrm{klas}}$	$\mathrm{N}_{\mathrm{klas}}$
$\mathrm{P}_{\mathrm{exp}}$	TP	FN
$\mathrm{N}_{\mathrm{exp}}$	FP	TN

Czułość klasyfikatora $Sens = \frac{TP}{TP+FN}$

Specyficzność klasyfikatora $Spec = \frac{TN}{TN+FP}$

Krzywa ROC
ang. Receiver Operating Characteristic





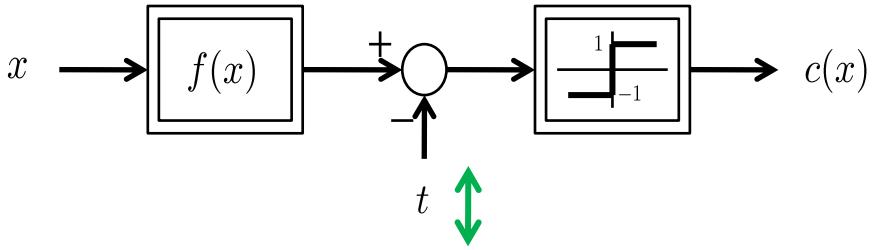
	Decyzja klasyfikatora	
Wiedza eksperta	$\mathrm{P}_{\mathrm{klas}}$	$\mathrm{N}_{\mathrm{klas}}$
$\mathrm{P}_{\mathrm{exp}}$	TP	FN
$\mathrm{N}_{\mathrm{exp}}$	FP	TN

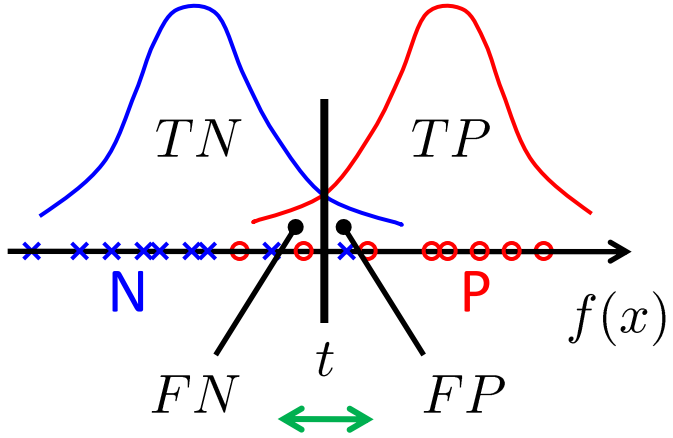
Czułość klasyfikatora $Sens = \frac{TP}{TP+FN}$

Specyficzność klasyfikatora $Spec = \frac{TN}{TN+FP}$

Krzywa ROC

ang. Receiver Oparating Characteristic





	Decyzja klasyfikatora	
Wiedza eksperta	$\mathrm{P}_{\mathrm{klas}}$	$\mathrm{N}_{\mathrm{klas}}$
$\mathrm{P}_{\mathrm{exp}}$	TP	FN
$\mathrm{N}_{\mathrm{exp}}$	FP	TN

Czułość klasyfikatora

$$Sens = \frac{TP}{TP+FN}$$

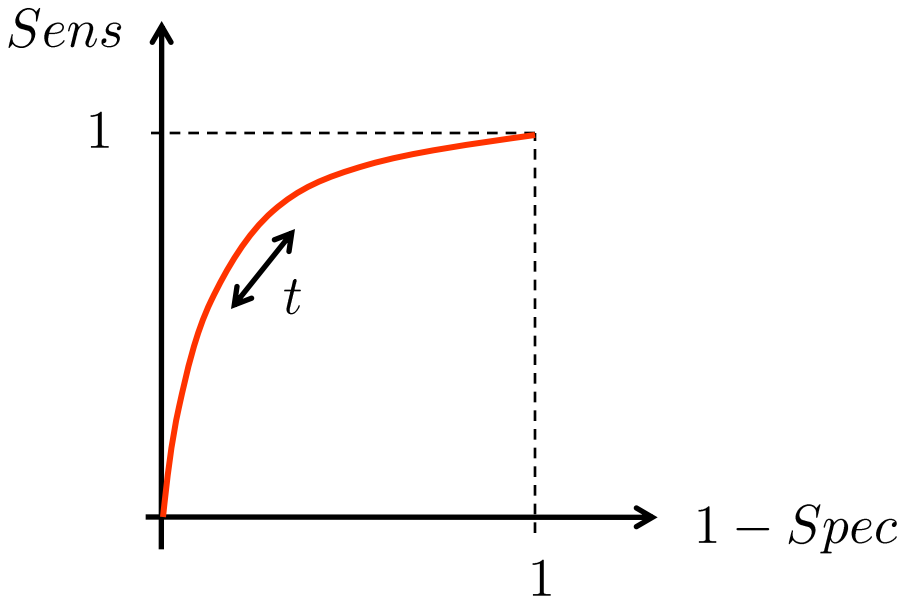
Specyficzność klasyfikatora

$$Spec = \frac{TN}{TN+FP}$$

Krzywa ROC
ang. Receiver Operating Characteristic

Krzywa ROC jest wykresem zależności $Sens$ od $1 - Spec$

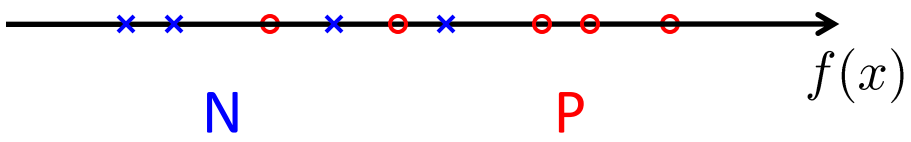
Parametrem tej krzywej jest próg t



Krzywa ROC

ang. Receiver Operating Characteristic

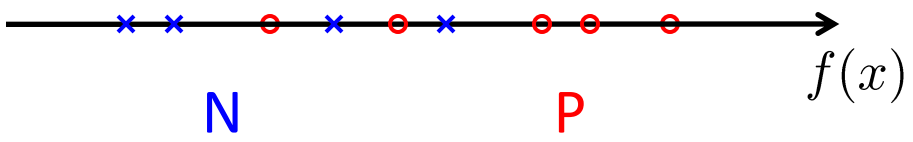
Przykład



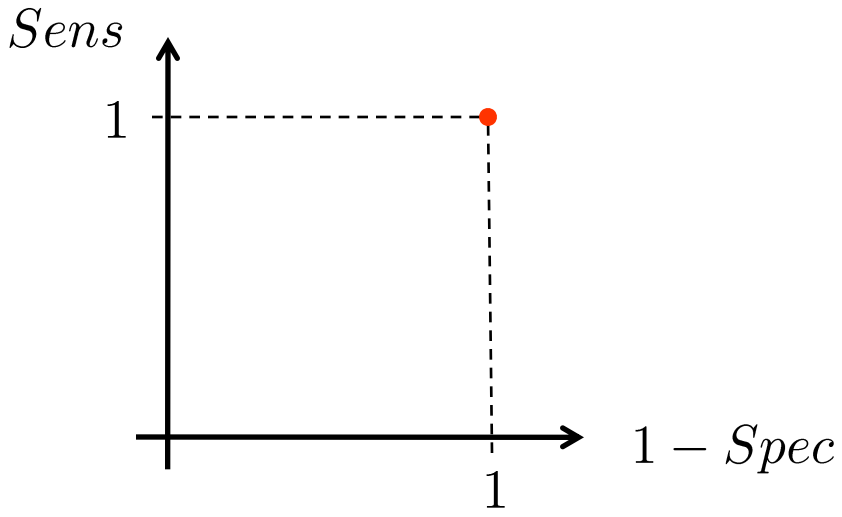
Krzywa ROC

ang. Receiver Operating Characteristic

Przykład



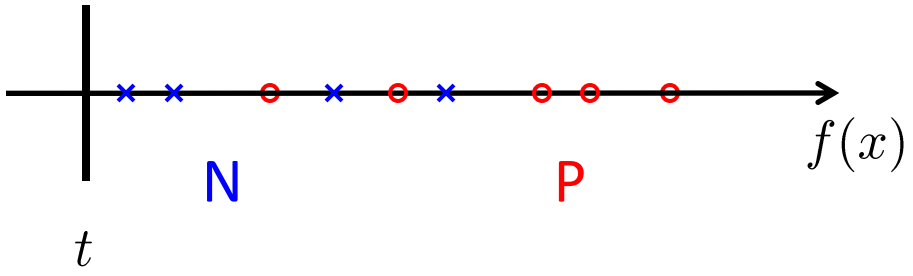
$Sens$	
$Spec$	
$1 - Spec$	



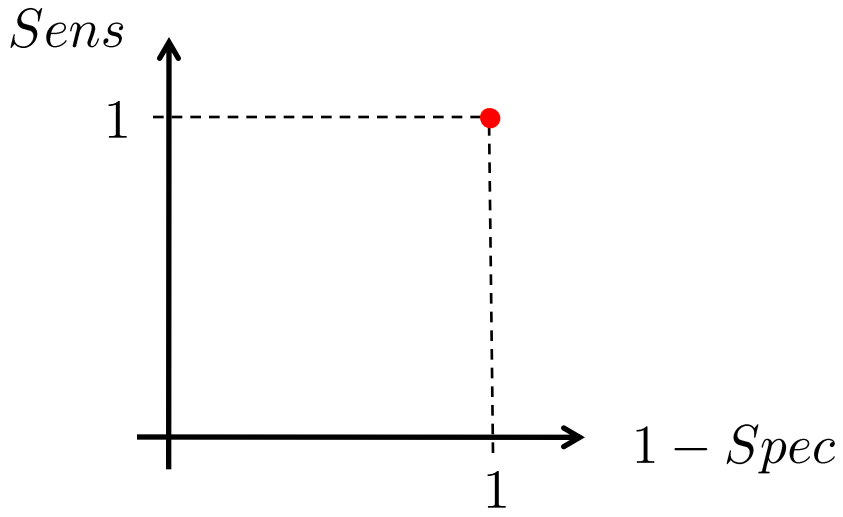
Krzywa ROC

ang. Receiver Operating Characteristic

Przykład



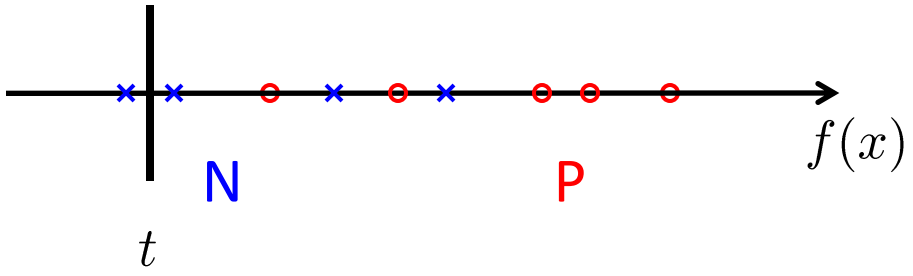
$Sens$	1
$Spec$	0
$1 - Spec$	1



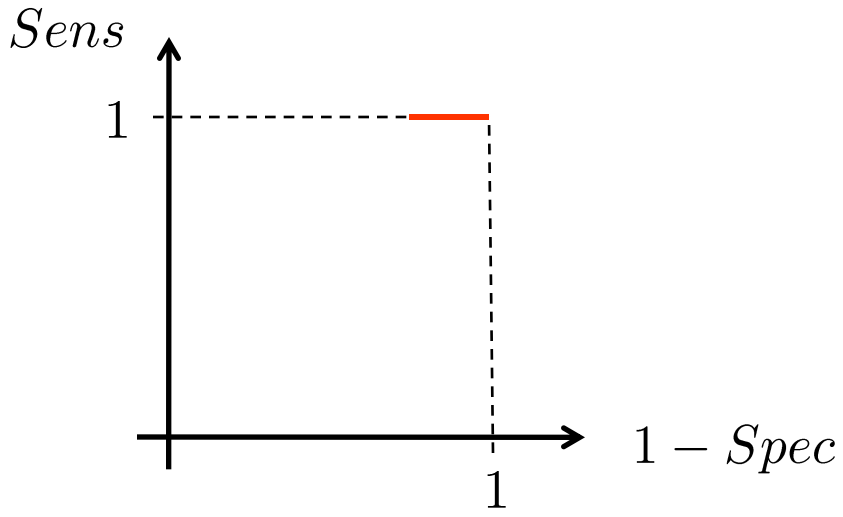
Krzywa ROC

ang. Receiver Operating Characteristic

Przykład



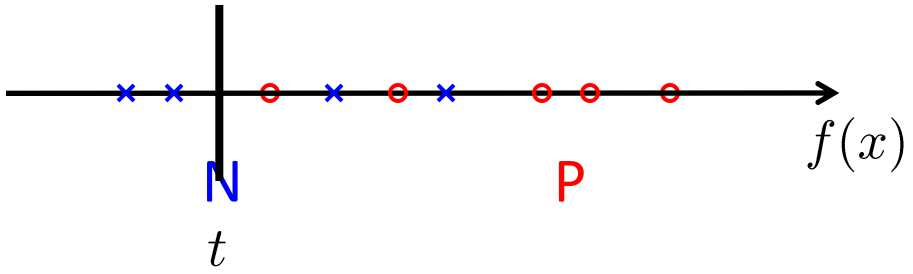
$Sens$	1	1
$Spec$	0	0,25
$1 - Spec$	1	0,75



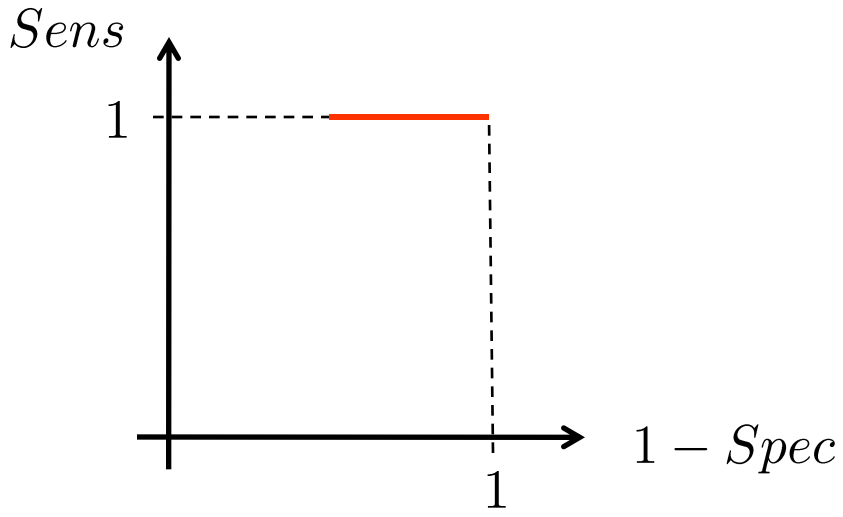
Krzywa ROC

ang. Receiver Operating Characteristic

Przykład



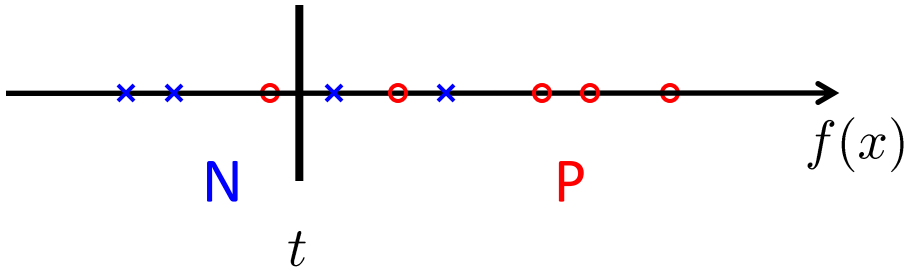
$Sens$	1	1	1
$Spec$	0	0,25	0,5
$1 - Spec$	1	0,75	0,5



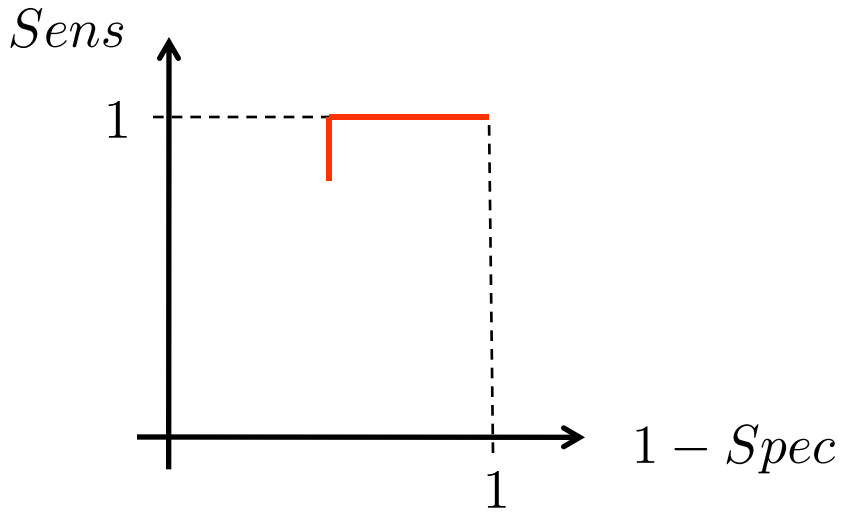
Krzywa ROC

ang. Receiver Operating Characteristic

Przykład



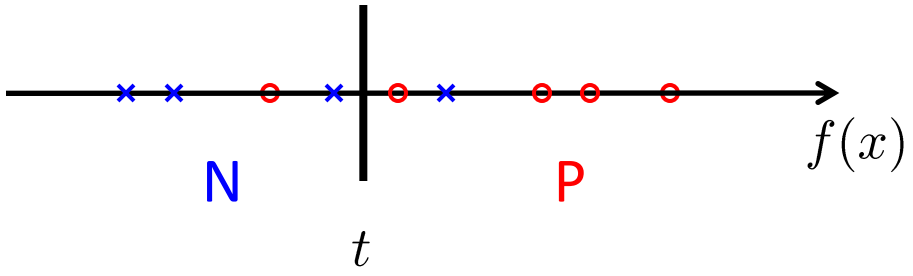
$Sens$	1	1	1	0,8
$Spec$	0	0,25	0,5	0,5
$1 - Spec$	1	0,75	0,5	0,5



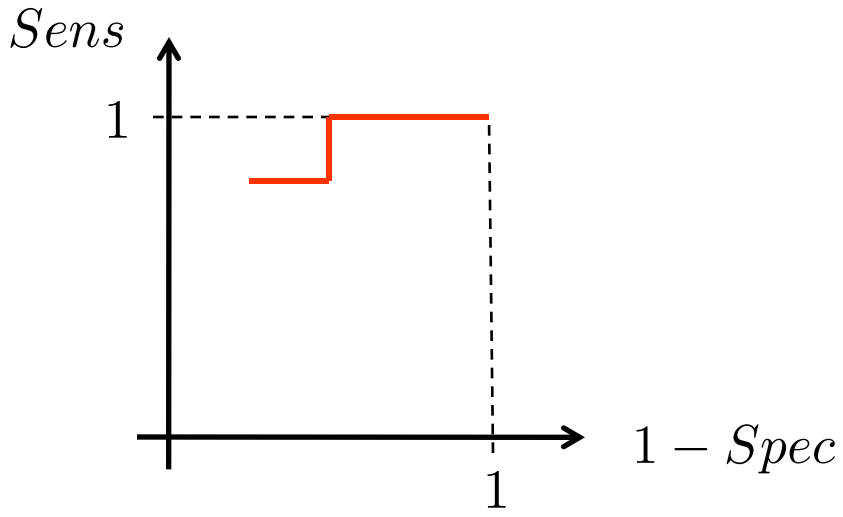
Krzywa ROC

ang. Receiver Operating Characteristic

Przykład



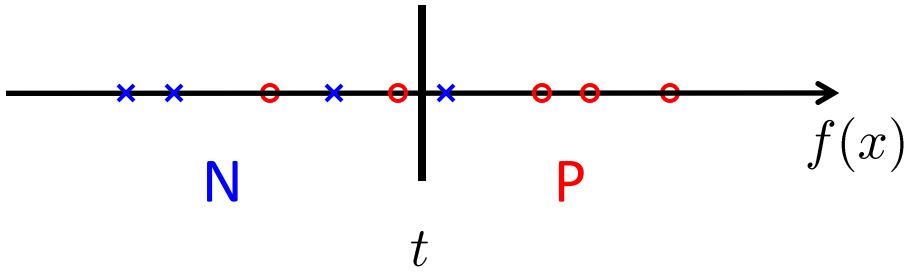
$Sens$	1	1	1	0,8	0,8
$Spec$	0	0,25	0,5	0,5	0,75
$1 - Spec$	1	0,75	0,5	0,5	0,25



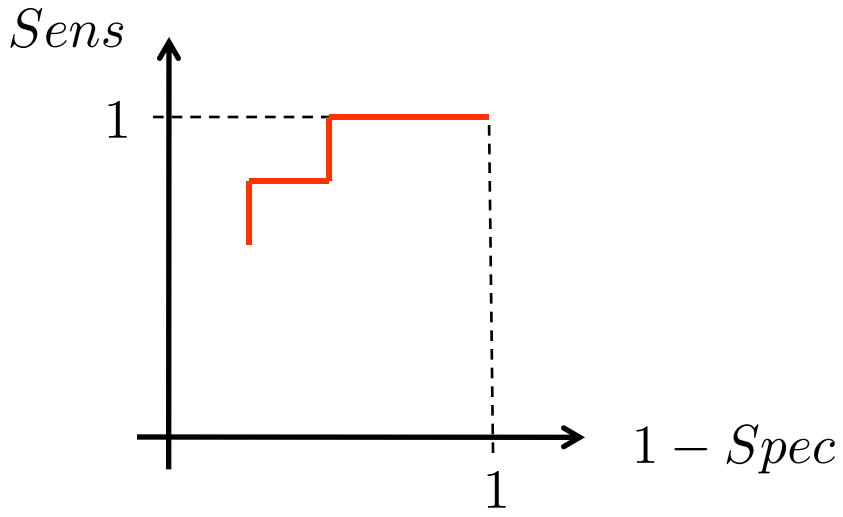
Krzywa ROC

ang. Receiver Operating Characteristic

Przykład



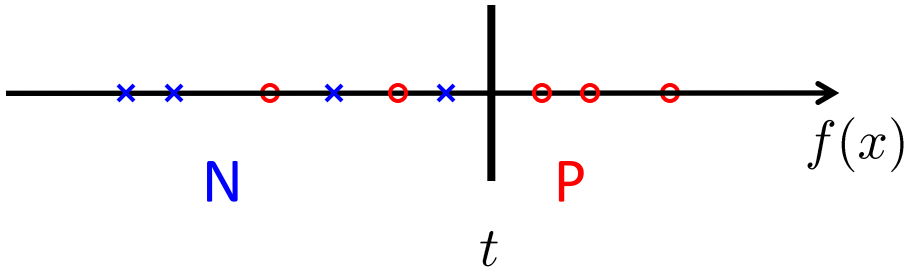
$Sens$	1	1	1	0,8	0,8	0,6
$Spec$	0	0,25	0,5	0,5	0,75	0,75
$1 - Spec$	1	0,75	0,5	0,5	0,25	0,25



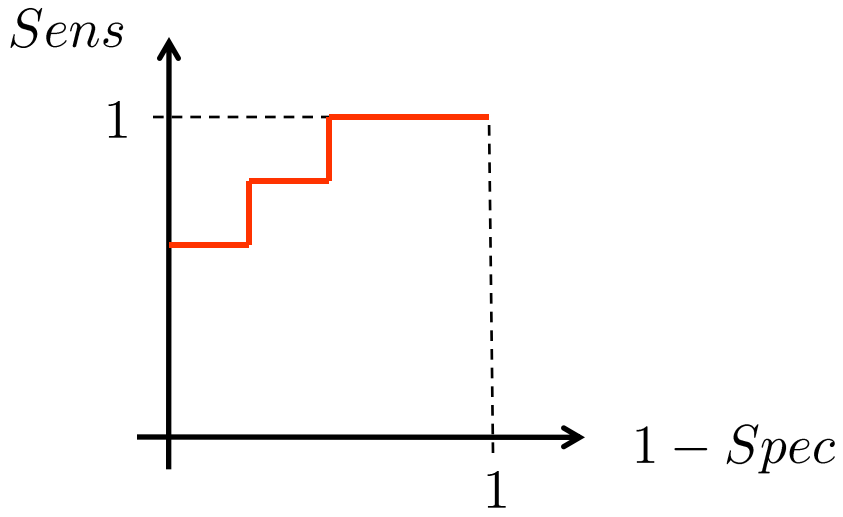
Krzywa ROC

ang. Receiver Operating Characteristic

Przykład



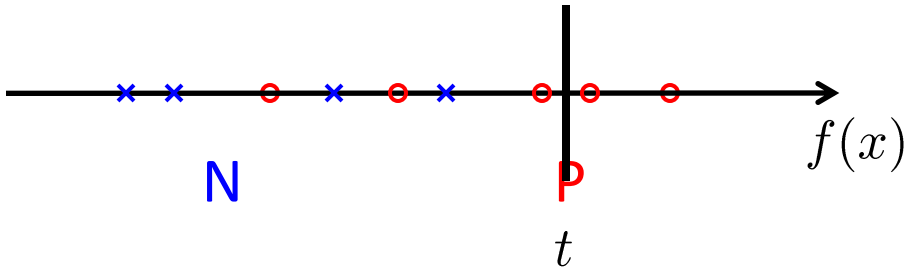
$Sens$	1	1	1	0,8	0,8	0,6	0,6
$Spec$	0	0,25	0,5	0,5	0,75	0,75	1
$1 - Spec$	1	0,75	0,5	0,5	0,25	0,25	0



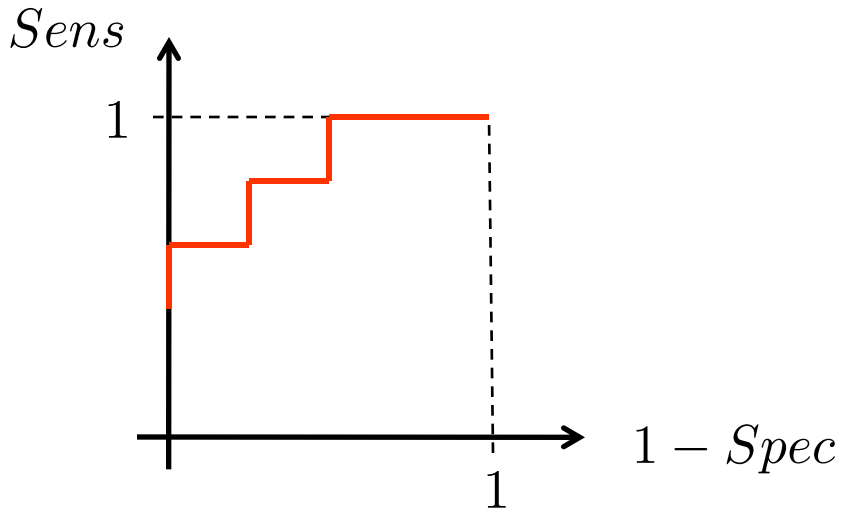
Krzywa ROC

ang. Receiver Operating Characteristic

Przykład

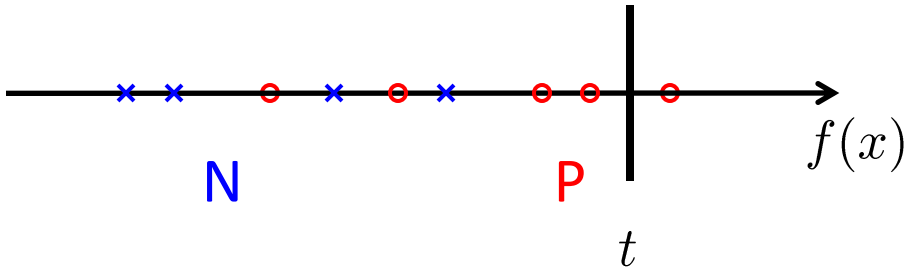


$Sens$	1	1	1	0,8	0,8	0,6	0,6	0,4
$Spec$	0	0,25	0,5	0,5	0,75	0,75	1	1
$1 - Spec$	1	0,75	0,5	0,5	0,25	0,25	0	0

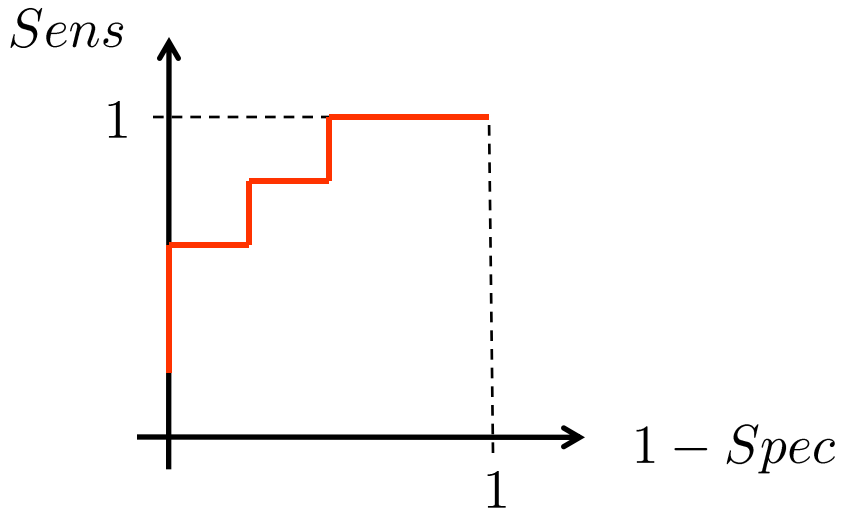


Krzywa ROC
ang. Receiver Operating Characteristic

Przykład



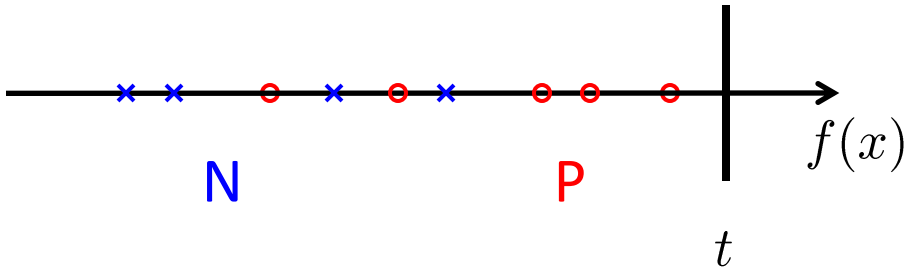
$Sens$	1	1	1	0,8	0,8	0,6	0,6	0,4	0,2
$Spec$	0	0,25	0,5	0,5	0,75	0,75	1	1	1
$1 - Spec$	1	0,75	0,5	0,5	0,25	0,25	0	0	0



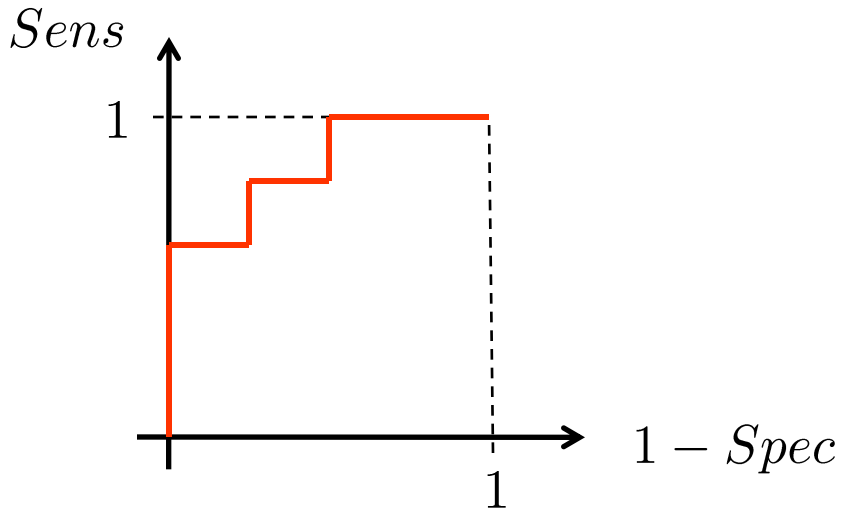
Krzywa ROC

ang. Receiver Operating Characteristic

Przykład

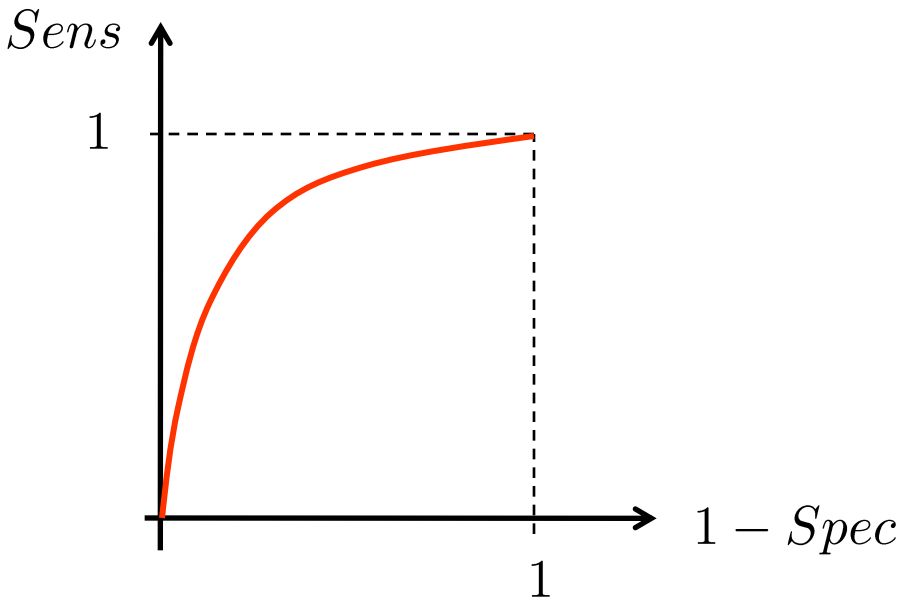


$Sens$	1	1	1	0,8	0,8	0,6	0,6	0,4	0,2	0
$Spec$	0	0,25	0,5	0,5	0,75	0,75	1	1	1	1
$1 - Spec$	1	0,75	0,5	0,5	0,25	0,25	0	0	0	0

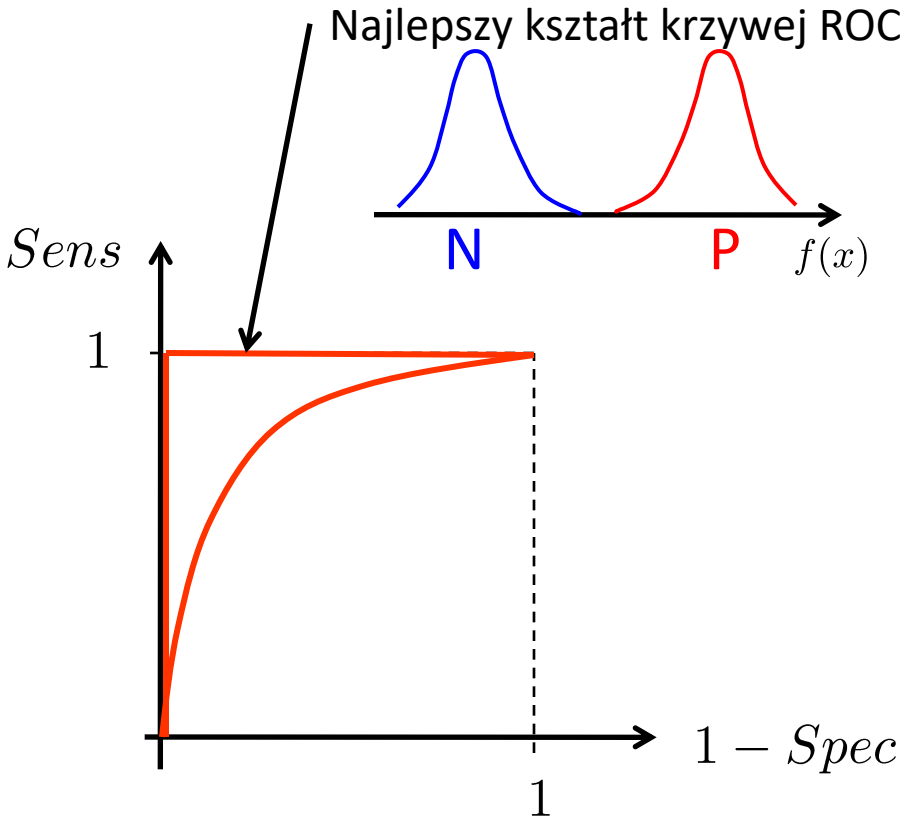


Krzywa ROC

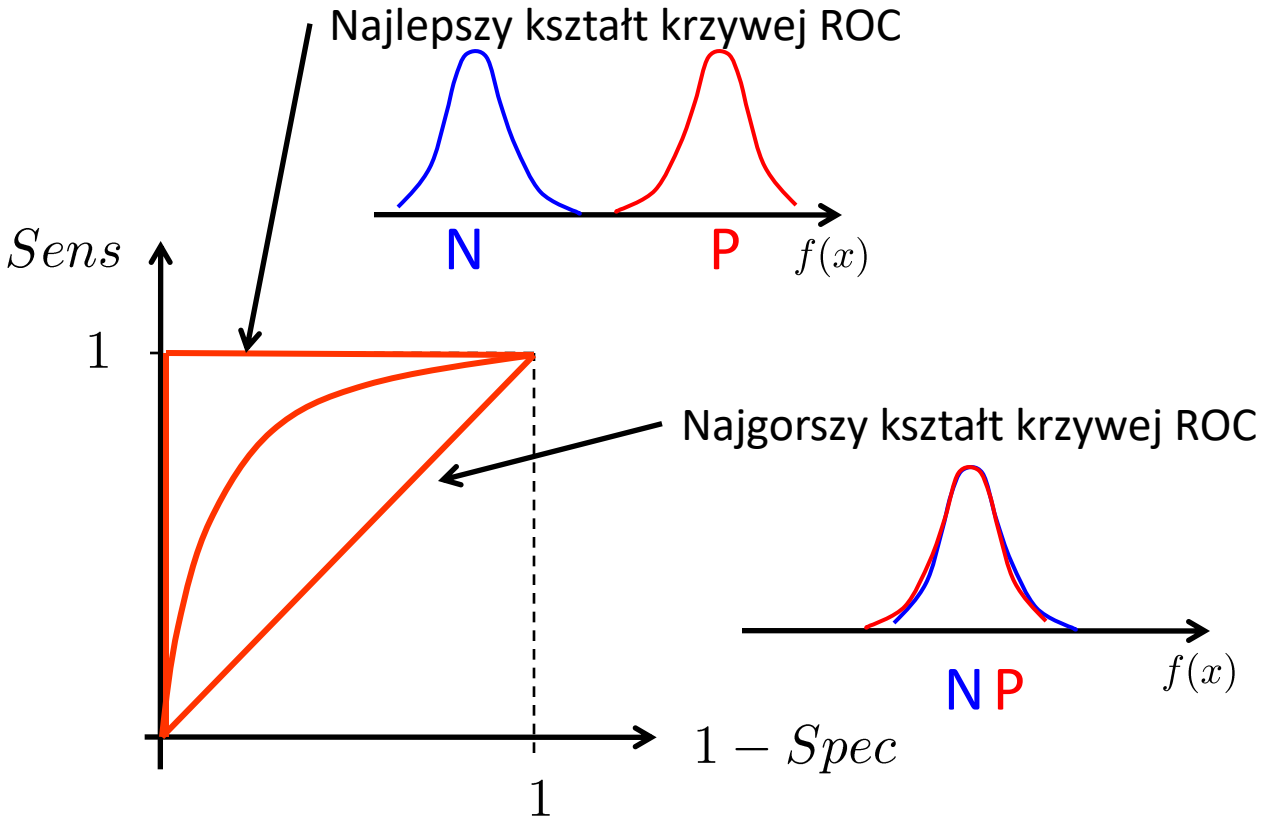
ang. Receiver Operating Characteristic



Krzywa ROC
ang. Receiver Operating Characteristic

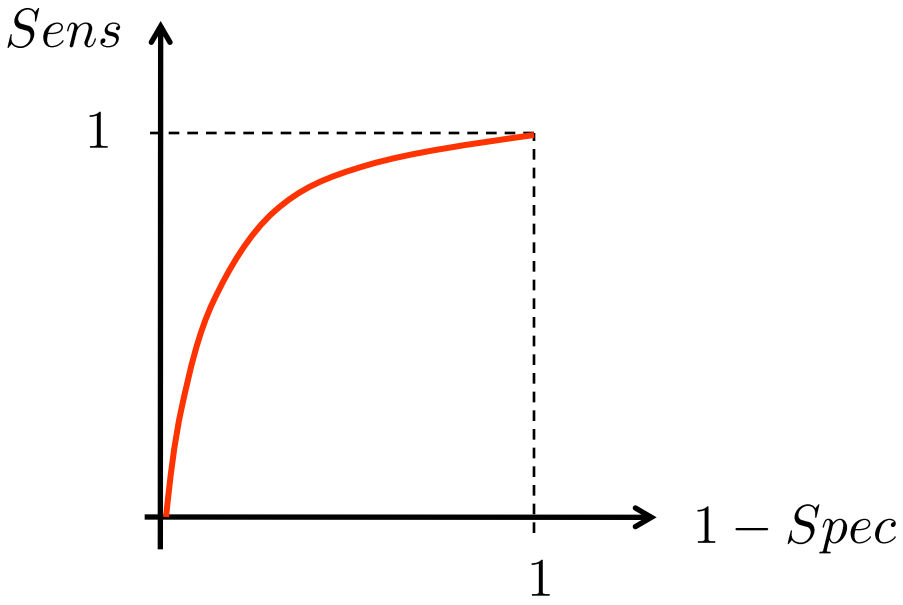


Krzywa ROC
ang. Receiver Operating Characteristic



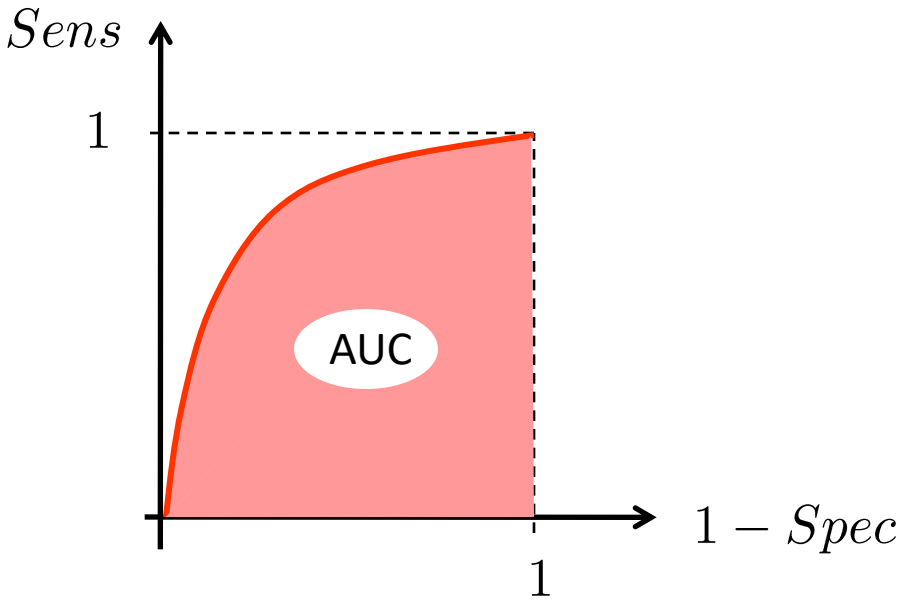
Pole pod krzywą ROC

ang. Area under curve (AUC)

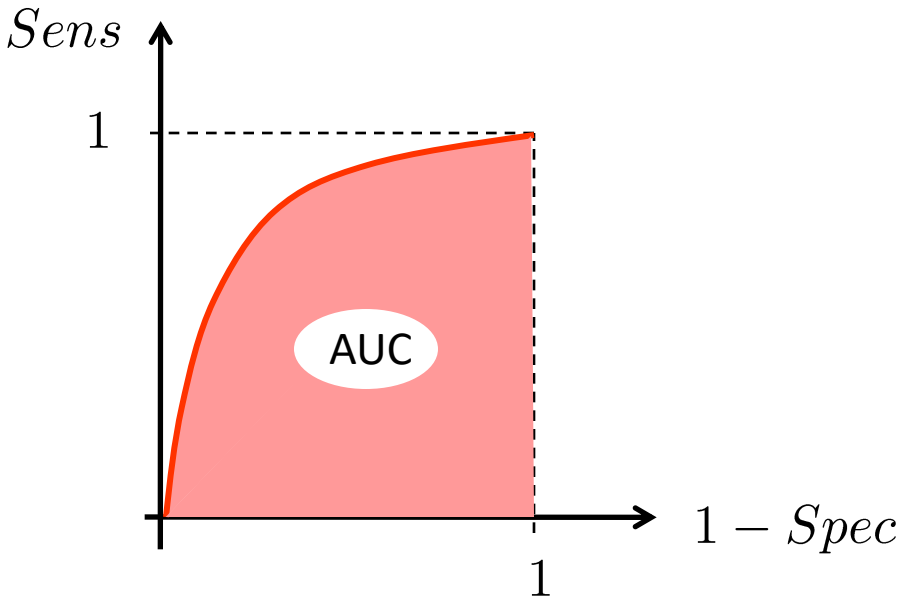


Pole pod krzywą ROC

ang. Area under curve (AUC)



Pole pod krzywą ROC
ang. Area under curve (AUC)



Wskaźnik jakości AUC, pomimo że jest najtrudniejszy do obliczenia, jest uznawany za najlepszy liczbowy wskaźnik jakości klasyfikacji.
Pole pod krzywą ROC wyraża prawdopodobieństwo, że dwa losowo wybrane wzorce z różnych klas mogą być prawidłowo zaklasyfikowane przez odpowiedni dobór progu.
AUC=1 oznacza klasyfikator idealny, a AUC=0,5 wskazuje na całkowity brak korelacji odpowiedzi klasyfikatora z prawdziwą klasą.
W istocie AUC odpowiada nieparametrycznemu testowi U Manna-Whitneya wykonanemu na wartościach funkcji klasyfikującej.